

W3 Day3

API (Application Programming Interface)

It is a mechanism that allows different software system to interact with each other.

Components of API

1. API Endpoint

It is a specific address (URL) where the client sends a request to access or modify something. for eg:-

<https://google.com/api/review/101>

2. HTTP Methods

It tell the server what you want to do. ~~for~~ common methods are

Methods	Meaning
GET	Get / read Data
POST	Create something
PUT	Replace / update something
PATCH	Partially Replacement / Update
DELETE	Remove something

3. Headers

It provide extra information about request or response

4. Parameters / Query

They are additional values that tell the server exactly what data you want or how you want the request processed. For eg
 /api/restaurant?KTM=loc & cat=pizza

Parameters

5. Request Body

The request body contains the actual data that the client sends to the server. For eg

{

"restaurant Id": 101,

"Food": "Chicken Pizza";

"quantity": 2;

"address": "Kathmandu"

}

6. Response

It is the information the server sends back after processing the client's request. For eg:

{

"orderId": 5001;

"Status": "Confirmed";

"Message": "Your order has been placed successfully"

}

7. HTTP Status Code

200 - OK

201 - Created

400 - ~~Bad~~ Bad Request

401 - Unauthorized

403 - Forbidden

404 - Not found

505 - Internal Server Error

Types of API Architectures

1. REST API (Representational State Transfer)

It uses HTTP methods to enable communication between client & server (JSON)

2. SOAP (Simple Object Access Protocol)

It uses XML based messages and follows standardized rules for communication between application. (Follows a strict rules used in banking) (WS-Security is used encryption, digital signature).

3. Graph QL

It is a query language that allows the client to request exactly the data it needs. for eg: The Real world Social Media App that want your name, PP, and email

request

```

{
  query {
    user (id: 101) {
      name
      profile picture
      email
    }
  }
}

```

Response

```

{
  "data": {
    "user": {
      "name": "Ram",
      "pp": "ram.jpg",
      "email": "ram.gmail.com"
    }
  }
}

```

4. gRPC (google Remote Procedure call)

It is a high performance framework that allows one application / service to directly call a function on another service. It can combine multiple server like Order service, Payment service, ProcessPaymentOrderId,

Types of API

1. Public API

It is available to external developers or the public.

2. Private API

It is an API used inside an organization to allow its own applications and services to communicate.

3. Partner API

It is an API shared with specific external companies or business partners.

4. Composite API

It combines multiple API operations or services into one API request.

FAST API

It is a python framework used to build APIs quickly and easily.

Uvicorn -

It is a server used to run ~~fast~~ Fast API application.

```
defining a path in fast api
@app.get("/")
def root():
```

Principles while designing Rest API

i) Uniform Interface

It should follow a standardized structure like GET /user/5, Delete /user/5.

ii) Client Server

The UI (client) and DS (server) should be able to work independently.

iii) Stateless

Each request must be full filling with each for the process without relying on past sessions.

iv) Cacheable

responses must be caches that allowing clients to reuse data for future.

v) Layered System.

The architecture is built in hierarchical layers.

API Key

A simple, long lived string (secret) passed by the client in the request header.

Bearer Token

It is a security token used in HTTP's auth^{authentication}. It gives access to the bearer of this token.

JWT (JSON Web Token)

It is a compact and secure method for transmitting information between parties as a JSON object.

JWT Structure

i) Header

It contains metadata about the token like algorithm (HS256), token type "JWT", "Auth".

ii) Payload

It contains the info about the user with some additional information.

- "iss" = Issuer
- "exp" = Expiration
- "iat" = Issued At

iii) Signature

It ensure token integrity, that the header, payload & ~~secret~~ secret keys are not being tempered sign is ~~not~~ generated using them. so they will be unique.

OAuth

It allows secure third-party access without sharing user credentials.

Idempotency

Same input always give same output

API Versioning

~~The way~~ It is a way of making API in such way the updates don't break the existing client app. for eg

api.example.com/v1/clients

api.example.com/v2/clients.

Pagination & Rate Limiting.

Pagination splits ~~the~~ the data into smaller manageable chunks.

Rate limiting restricts how many requests a client can make to an API within a specific time.